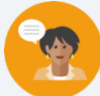


1.5 Solving Mathematical and Real-World Problems with Equations

Lesson Introduction

 Benchmark	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.		 Learning Targets	- I can solve two-step equations involving the distributive property. - I can solve two-step equations within a real-world context. - I can relate the solution of an equation representing a real-world context to the context.	
 Benchmark Coherence	Building On			Working Toward	
	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.			MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.	
 Essential Questions	- How do you apply the distributive property to solve two-step equations? - How do you relate the solution of an equation representing a real-world context to the context? - How do you solve two-step equations?				
 Lesson Narrative	In this lesson, students review the distributive property and apply it to solve two-step equations. Students learn that they can solve equations of the form $p(x + r) = q$ either by using the distributive property or the division property of equality. This lesson guides students through writing equations in the form $px + q = r$ and $p(x + q) = r$. The lesson is scaffolded throughout to help build student thinking about real-world contexts and the equations that model them. Students also relate given equations to the real-world contexts they represent. Finally, students solve and interpret solutions based on the real-world context.				
 Component Summary	1.5.1 Warm-Up (~5 minutes)	Describe a pattern (SE p. 30)	1.5.4 Collaboration (10 min.)	Match equation to bar diagrams and real-world contexts (SE p. 33)	
	1.5.2 Refresher (5 min.)	Review distributive property (SE p. 30)	1.5.5 Guided Instruction (5 min.)	Practice relating equations to their contexts (SE p. 34)	
	1.5.3 Guided Instruction (20 min.)	Learn how to solve equations involving the distributive property (SE p. 31)	1.5.6 Try It (5 min.)	Independent practice relating equations to their contexts (SE p. 35)	
	1.5.7 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 36)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Reciprocal - Addition, subtraction, multiplication, and division - Properties of equality - Distributive property		- Calculators (Optional) Chart paper and markers (Optional) Cups to demonstrate for question 2a in 1.5.4.		Consider creating and displaying an anchor chart with the properties of equality from previous lessons to aid students in solving two-step equations.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recognize the equivalence of $p(x + r)$ and $(x + r)p$. - Students may add or subtract the coefficient of the variable instead of dividing both sides of the equation by the coefficient. - Students may not distribute to both terms inside of the parentheses. - Students may not perform the inverse operations on both sides of the equation. - Students may struggle relating the solution to the real-world context.	- Model underlining key information in real-world problems. Then, explain how to relate the key information to the equation. - Ask students to identify something that varies and something that remains constant in the real world. - Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. - Determine if your students need the support of all or part of the Refresher. - If time is an issue, then consider using 1.5.4 Collaboration question 2b as part of 1.5.5 Guided Instruction. - If time is an issue, then consider assigning Try It for homework.
 Differentiate Instruction	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 1.5.4 Collaboration students match equations to a bar diagram and real-world contexts. Ask students to initially complete the problems independently and then discuss their work. These discussions will help students gain confidence and understanding.	MA.K12.MTR.7.1: World Contexts In 1.5.6 Try It, students solve two similar equations and determine which equation represents a real-world problem. Students are asked to explain what the solution to the equation that correctly models this situation means in the context. Ask students whether the solutions make sense given the context. Solving real-world problems increases student interest and appreciation of the utility of mathematics.
Lesson Notes:		

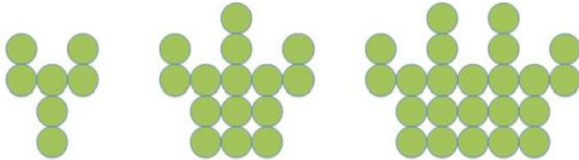
1.5.1 Warm-Up: Number Patterns

Component Overview: Students explore, describe, and make predictions based on a visual pattern. Consider advising students to make a two-column table with the step number in column one and the number of circles in column two. This component provides a visual entry point for students.



1.5.1 Warm-Up: Number Patterns

A pattern is shown.



- Describe how the pattern is changing in each step.
Answers will vary. In each step you are adding a column of 3 circles and a column of 5 circles to the interior.
- How many circles will be in the next step?
31 circles



Ask students to define a variable and generalize the pattern with an algebraic expression. Provide students with additional visual patterns and/or ask the student to create their own visual patterns.

1.5.2 Refresher: The Distributive Property

Component Overview: Students review when and how to use the distributive property. Students practice applying the distributive property to three different expressions.



1.5.2 Refresher: The Distributive Property

1. Several expressions are shown.

$$(-2 \cdot x) + 3$$

$$(x + 3)(-2)$$

$$-2x - 6$$

$$(-2) \cdot x + (-2) \cdot 3$$

$$(-2 + x)3$$

- a. Circle the expressions that are equivalent to $-2(x + 3)$.
- b. Use substitution to show that each of the expressions chosen is equivalent to $-2(x + 3)$.

$$\text{Let } x = 1$$

$$-2(1 + 3) = -8$$

$$(1 + 3)(-2) = -8$$

$$-2(1) - 6 = -8$$

$$(-2) \cdot 1 + (-2) \cdot 3 = -8$$

2. Rewrite each expression by applying the distributive property.

a. $(-x + 6)4$
 $-4x + 24$

b. $-3.5(-7 + 2y)$
 $24.5 - 7y$

c. $\frac{7}{8}\left(16m - \frac{4}{5}\right)$
 $14m - \frac{7}{10}$



Recognizing the expressions are equivalent may be difficult for students. Help students see the structure in the distributive property by demonstrating how $p(x + r)$ and $(x + r)p$ are equivalent based on the commutative property of multiplication.

Understanding equivalency of expressions and equations empowers students to think flexibly. Consider incorporating different equivalent expressions and equations as you work with students to model flexible thinking.



To provide a visual entry-point, encourage students to draw arrows to each term inside the parenthesis.

1.5.3 Guided Instruction: Solving Equations Involving the Distributive Property

Component Overview: Students learn how to solve equations involving the distributive property. Students learn that they can either use the distributive property or the division property of equality to solve the equations.



1.5.3 Guided Instruction: Solving Equations Involving the Distributive Property

Equations can be solved in more than one way. When an equation is written in the form $a(b + c) = d$, it can be solved by either distributing a first or by dividing both sides the equation by a first.

1. The equation $65 = 8(9 - p)$ can be solved using either method.
 - a. Solve the equation $65 = 8(9 - p)$ using each method.

Method 1 Apply the distributive property first	Method 2 Divide both sides by the factor being distributed first
$ \begin{aligned} 65 &= 8(9 - p) \\ 65 &= 72 - 8p \\ 65 - 72 &= 72 - 72 - 8p \\ -7 &= -8p \\ \frac{-7}{-8} &= \frac{-8p}{-8} \\ \frac{7}{8} &= p \end{aligned} $	$ \begin{aligned} 65 &= 8(9 - p) \\ \frac{65}{8} &= \frac{8(9 - p)}{8} \\ 8\frac{1}{8} &= 9 - p \\ 8\frac{1}{8} - 9 &= 9 - 9 - p \\ -\frac{7}{8} &= -p \\ -1 * \left(-\frac{7}{8}\right) &= -1 * (-p) \\ \frac{7}{8} &= p \end{aligned} $

- b. Verify the solution is correct using substitution.

$$\begin{aligned}
 65 &= 8(9 - p) \\
 65 &= 8\left(9 - \left(\frac{7}{8}\right)\right) \\
 65 &= 8\left(8\frac{1}{8}\right) \\
 65 &= 65
 \end{aligned}$$



If students write $65 = 72 - p$ remind them to multiply 8 by both terms in the parenthesis. Students may feel more comfortable solving the equation if it is rewritten as $8(9 - p) = 65$, because the variable is on the left side.



Students should be encouraged to utilize methods of solving equations that make sense for them. Although there are common methods that teachers consider to be easier, this may not be a method that a student considers easier. There are also students who do not like to be presented with options. Help these students embrace equivalency even if they would prefer not to consider alternate methods.

1.5.3 Guided Instruction: Solving Equations Involving the Distributive Property (continued)

- c. Discuss each method of solving the equation with your partner. Then, explain which method you preferred for solving this equation.

Answers will vary.

2. With your partner, decide who is Partner A and who is Partner B.

- a. Solve the equation $8\frac{1}{2} = \frac{2}{3}(x + 6)$ using the method indicated.

Partner A Apply the distributive property first	Partner B Divide both sides by the factor being distributed first
$8\frac{1}{2} = \frac{2}{3}(x + 6)$ $8\frac{1}{2} = \frac{2}{3}x + 4$ $8\frac{1}{2} - 4 = \frac{2}{3}x + 4 - 4$ $4\frac{1}{2} = \frac{2}{3}x$ $4\frac{1}{2} \cdot \frac{3}{2} = \frac{2}{2} \cdot \frac{3}{3}x$ $6\frac{3}{4} = x$	$8\frac{1}{2} = \frac{2}{3}(x + 6)$ $\frac{3}{2} \cdot 8\frac{1}{2} = \frac{3}{2} \cdot \frac{2}{3}(x + 6)$ $12\frac{3}{4} = x + 6$ $12\frac{3}{4} - 6 = x + 6 - 6$ $6\frac{3}{4} = x$

- b. Share your solution and your work with your partner. Correct any errors, if necessary. Then, write your partner's method in the space provided once both partners agree the work is correct.

Answers will vary.

- c. Explain which method you preferred for solving this equation.

Answers will vary.



Consider challenging students who finish early to consider Partner C's work.
Partner C

$$8\frac{1}{2} = \frac{2}{3}(x + 6)$$

$$\frac{8\frac{1}{2}}{\frac{2}{3}} = \frac{\frac{2}{3}(x + 6)}{\frac{2}{3}}$$



Think-Pair-Share

Have students initially complete the question independently and then discuss their work with their partners. The discussions will build student confidence and understanding.

1.5.3 Guided Instruction: Solving Equations Involving the Distributive Property (continued)

3. The perimeter of an equilateral triangle is 41.46 inches. If the length of each side is $(q - 5.7)$, the equation $3(q - 5.7) = 41.46$ can be used to find the value of q .

- a. What is the value of q ?

$$\begin{aligned} 3(q - 5.7) &= 41.46 \\ 3q - 17.1 &= 41.46 \\ 3q - 17.1 + 17.1 &= 41.46 + 17.1 \\ 3q &= 58.56 \\ \frac{3q}{3} &= \frac{58.56}{3} \\ q &= 19.52 \end{aligned}$$

- b. The length of each side of the triangle is 13.82 inches because $41.46 \div 3 = 13.82$. Explain how the solution to the equation $3(q - 5.7) = 41.46$ can be used to show this as well.

The length of one side is represented by $q - 5.7$, and q is equal to 19.52. The length of one side of the triangle is $19.52 - 5.7 = 13.82$. The side length can be multiplied by 3 to find the perimeter, 41.46.



Consider challenging students who finish early to solve the problems using both the distributive property and the division property of equality.



What are the features of an equilateral triangle? Which part of the equation tells us the length of one side of the triangle? How do you determine the perimeter of a triangle?

1.5.4 Collaboration: Matching Equations to Diagrams and Contexts

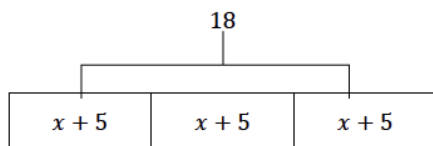
Component Overview: Students match equations to bar diagrams and real-world contexts. Students explore each part of the equation to determine the proper match. This component lays the foundation for subsequent discussions of unknowns, variables, and constants.



1.5.4 Collaboration: Matching Equations to Diagrams and Contexts

Equations can be represented using bar diagrams.

1. A bar diagram is shown.



What do you notice about the equations that represent the bar diagram? What do you notice about the equations that do not match the bar diagram? How many $x + 5$ expressions are there?

- a. Select all the equations that match the diagram.

- ☐ $x + 5 = 18$
- ☐ $18 \div 3 = x + 5$
- ☐ $3(x + 5) = 18$
- ☐ $x + 5 = \frac{1}{3} \cdot 18$
- ☐ $3x + 5 = 18$

- b. Compare your selections with your partner. Resolve any differences if necessary.

Answers will vary.

- c. With your partner, choose one of the equations that can be modeled by the bar diagram and explain why it matches the diagram.

Answers will vary.

$3(x + 5) = 18$ this is equivalent because there are 3 groups of $x + 5$ which altogether equal 18.

- d. Explain why $x + 5 = 18$ does not match the bar diagram.

$x + 5 = 18$ does not match the bar diagram because the bar diagram show 3 groups of $x + 5$, not 1 group of $x + 5$.

1.5.4 Collaboration: Matching Equations to Diagrams and Contexts (continued)

Equations can be used to represent real-world contexts.

2. Three equations are shown.

Equation A	Equation B	Equation C
$\frac{1}{4} + 4x = 8$	$4 + \frac{1}{4}x = 8$	$8x + \frac{1}{4} = 4$

- a. Work with your partner to determine which equation matches each context. Then, write each equation in the appropriate row.

Context	Equation
A stack of nested paper cups is 8 inches tall. The first cup is 4 inches tall and each of the rest of the cups in the stack adds $\frac{1}{4}$ inch to the height of the stack.	<i>Equation B</i>
A baker uses 4 cups of flour. She uses $\frac{1}{4}$ cup to cover the counter with flour so the dough doesn't stick, and the rest to make 8 identical loaves of bread.	<i>Equation C</i>
Elena has an 8-foot piece of ribbon. She cuts off a piece that is $\frac{1}{4}$ of a foot long and cuts the remainder into four pieces of equal length.	<i>Equation A</i>

- b. With your partner, discuss what the variable x represents in each of the equations. Then, write the meaning of x in each equation based on the context it matches.

Equation	Meaning of x
A: $\frac{1}{4} + 4x = 8$	<i>x is the remaining length of each piece of ribbon after cutting off $\frac{1}{4}$ of a foot.</i>
B: $4 + \frac{1}{4}x = 8$	<i>x is the remaining number of cups after the first cup.</i>
C: $8x + \frac{1}{4} = 4$	<i>x is the amount of flour in each loaf of bread.</i>



Check that all students understand the contexts in the problems. For example, students may need clarification on the meaning of “nested paper cups.” Consider showing students images of the contexts to provide a visual entry point. You can also modify the contexts to appeal to your students’ interests.



Consider splitting students into groups and assigning each group one of the equations (jigsaw strategy). Ask students to show their work on chart paper, including the equation, word problem, and justification of why they match. Use a gallery walk to debrief the problems.



Before Guided Instruction, review student responses to question 2b as a class. Consider asking these questions:

- What does the variable represent in each equation?
- Will the value of the variable stay the same or change?
- How did you determine which equation matched which context?



1.5.5 Guided Instruction: Relating Equations to Their Contexts

Component Overview: Students relate equations to their real-world contexts. The problems scaffold students through the steps needed to write an equation based on a real-world problem.



1.5.5 Guided Instruction: Relating Equations to Their Contexts

1. Franco is saving money to take a trip to Cocoa Beach, Florida with his friends. Every week, he sets aside the same portion of his paycheck after taking out the \$325 he needs for his weekly expenses. After 11 weeks, Franco has \$489.94 saved for his trip.

- a. An equation can be created relating these quantities as described below.

$$\left(\begin{array}{c} \text{Number} \\ \text{of} \\ \text{weeks} \end{array} \right) \left(\begin{array}{c} \text{Amount} \\ \text{of} \\ \text{weekly} \\ \text{paycheck} \end{array} - \begin{array}{c} \text{Amount} \\ \text{of weekly} \\ \text{expenses} \end{array} \right) = \begin{array}{c} \text{Total} \\ \text{Saved} \end{array}$$

Which of these quantities is unknown?

The amount of Franco's weekly paycheck is unknown.

- b. Write the equation that represents this situation using the variable p for the unknown quantity.

$$11(p - 325) = 489.94$$

- c. What does $(p - 325)$ represent in the equation?

The amount that Franco saves each week.

- d. Solve the equation.

$$\frac{11(p-325)}{11} = \frac{489.94}{11}$$

$$p - 325 = 44.54$$

$$p - 325 + 325 = 44.54 + 325$$

$$p = 369.54$$

- e. Explain what the solution means in this context.

Franco earns a paycheck of \$369.54 each week.



Consider allowing students to work with a partner. Students may benefit from hearing each other's reasoning and checking each other's work. These discussions will also provide auditory and visual entry-points as students hear and see each other's strategies.



What part of the equation will stay the same? What part of the equation will change? Does the solution make sense in the context?

1.5.6 Try It: Relating Equations to Their Contexts

Component Overview: Students independently solve two-step equations and determine which equation matches a real-world context. Students also explain what the solution means in the context of the problem.



1.5.6 Try It: Relating Equations to Their Contexts

Work with your partner to complete the following.

1. Two equations are shown.

a. Solve both equations. Show your work.

Equation A	Equation B
$46.92 = 6(y - 1.75)$ $\frac{46.92}{6} = \frac{6(y - 1.75)}{6}$ $7.82 = y - 1.75$ $7.82 + 1.75 = y - 1.75 + 1.75$ $9.57 = y$	$6y - 1.75 = 46.92$ $6y - 1.75 + 1.75 = 46.92 + 1.75$ $6y = 48.67$ $y \approx 8.11$

b. A family of 6 is going to the local fair. They have a coupon for \$1.75 off of each ticket. They pay \$46.92 for all their tickets. Explain which equation represents this situation.

Equation A represents this situation because the family is buying 6 tickets but receiving 1.75 off of each ticket, not just one ticket.

c. Explain what the solution to the equation that correctly models this situation means.

y represents the price of an individual ticket before the coupon is applied. So, each ticket cost \$9.57 before the coupon.



Ask students whether their solution makes sense given the context.



Solving real-world problems increases student interest and appreciation of the utility of mathematics. Consider extending the activity by challenging students to create real-world problems that can be represented by equations. Then, students can exchange real-world problems and write the related equations.

1.5.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



1.5.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.



For students who struggle with expressing themselves in writing, consider providing alternative options, such as using an audio recorder, speech-to-text software, or referring to problems in the lesson where they wrote and solved real-world two-step equations.